

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Digital Logic

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Compare analog systems with digital systems and list out the applications of digital systems. 7
b) Perform the conversion as indicated: 8
 - i. $(37.432)_{10} = (?)_2$
 - ii. $(24C5)_{16} = (\dots)_2 = (\dots)_{10}$
 - iii. $(58.43)_{10} = (?)_{\text{excess-3}}$
 - iv. 72532-3250 using 9's complement
2. a) What is weighted and non weighted code? Why gray code is called reflected code? 7
b) Define Literal. Prove the De-Morgan's first and second theorem with necessary diagrams and equations. 8
3. a) What are the Universal gates? Why are they called so? Explain. 7
b) A Boolean function $F(A,B,C,D) = \sum(0,1,2,3,10,13,14)$ and don't care condition $d(A,B,C,D) = \sum(4,7,12)$. Simplify it using K-Map and implement using NAND gate only. 8
4. a) Design Full Subtractor using two half adders and an OR gate. 7
b) Implement the following Boolean function 8
 $F(A,B,C,D) = \sum(0,1,2,5,6,11,13,14)$ using 8 to 1 Multiplexer and B,C,D as selection lines.

OR

A combinational circuit is defined by the functions $F_1(A,B,C) = \sum(3,5,6,7)$ and $F_2(A,B,C) = \sum(0,2,4,7)$. Implement the circuit with a PLA having three inputs, four product terms and two outputs.

5. a) Define Register. Design 6-bit SISO and PISO Register. 8

b) Design 3-bit UP/DOWN Counter.

7

OR

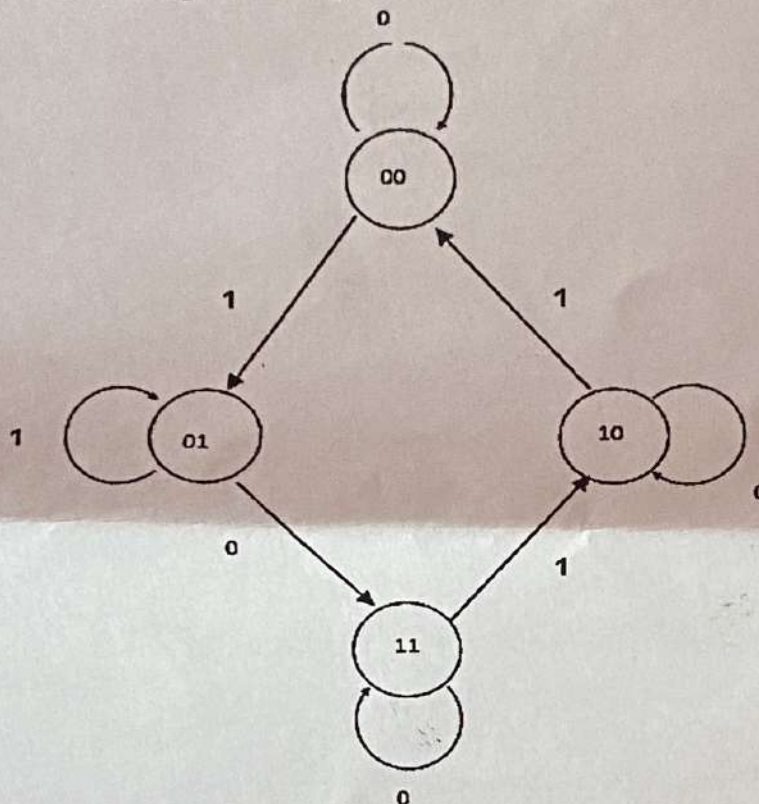
Explain Ring Counter in Detail.

6. a) Define flip flop. Explain the working of JK flip flop with its truth table and logic diagram. Also explain how it eliminates the problem encountered in SR flip flop.

7

OR

Realize the following state diagram into the circuit using T flipflop.



- b) Design a 4-bit arithmetic circuit which performs eight different arithmetic operations.

8

7. Write short notes on: (Any two)

2×5

- a) Status Register
- b) Accumulator
- c) RAM